

JOHNSON MUGARRA

Data Scientist and Research Consultant

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PROFESSIONAL SUMMARY

Data scientist and research consultant with an M.Sc. in Data Science and six years working with field and operational data across East Africa. At Enimiro I design studies (child labor, living income, pollination) and build the pipelines, dashboards, and yield models behind them. At the Center for Research and Communication I coordinate proposals and lead the technical work on evaluations covering design, sampling, tools, data quality, analysis, and reporting. Earlier roles cover vanilla value-chain forecasting with Makerere University and MAAIF, process evaluation at IPA, and field research for USAID SITES.

CORE SKILLS

- Statistical software: Stata, R, Python, SQL, SAS, SPSS, Excel
- Research and evaluation: survey design, piloting and translation, computer-assisted interviewing, key informant interviews, focus group discussions, transcription, process and impact evaluation, monitoring and evaluation frameworks, indicator tracking
- Data collection and quality: questionnaires, tablet-based collection, enumerator training and supervision, validation, backup and documentation
- Visualization and reporting: Power BI, Tableau, dashboards, reports for donors and management
- Data systems: SurveyCTO API, PostgreSQL, Google Sheets, ETL pipelines, GeoPandas
- Machine learning and AI: scikit-learn, time-series forecasting, neural networks and CNNs in TensorFlow and PyTorch, and daily use of AI tools for cleaning data, writing code, and drafting, with the output checked by hand
- Proposal development: concept note, strategy, design, methods and work plans, budgeting, quality assurance

WORK EXPERIENCE

Research and Data Consultant, November 2025 - October 2026

Enimiro Products Uganda Ltd, Kayunga, Uganda

- Coordinated the data team on collection, cleaning, and organization of operational data across farmer activities, crop production, sales, and sustainability.
- Maintained a Python pipeline moving SurveyCTO submissions into PostgreSQL raw, clean, and analysis tables, with scheduled exports to Google Sheets for dashboards.

- Designed studies from proposal to analysis, including the child labor study, the living income study against Anker benchmarks, pollination and maturity modules, and the M&E measurement plan.
- Matched GPS to district boundaries and maintained the farmer master with merge and change logs for audits.
- Built dashboards and reports tracking performance indicators for management and the board.
- Ran statistical and trend checks to flag problems and turned findings into training manuals and field guidance for staff.
- Built the data and M&E workstream: built the M&E measurement plan and monitoring toolkit (KPI dashboards, tier trackers, review packs) used across 19 district and managed the field-extension restructuring from concept note to roll-out.
- Supported crop and supply-chain research, updated survey tools, trained staff on data use, and documented methods for donor and certification reporting.

Technical Lead Consultant, January 2025 - Present

Center for Research and Communication (CRC), Remote

- Reviewed client requirements and evaluation criteria to set the proposal approach and coordinated specialist inputs against deadlines.
- Wrote and edited technical sections, including methods and work plans, and worked with finance to align technical approaches with compliant budgets.
- Led the technical work for the IRC MERP endline evaluation, covering design and tools, analysis and synthesis, and integration of qualitative and quantitative findings against the results framework.
- Led quantitative inputs for IRC baseline and perception studies: sampling, instrument testing, data-quality checks, and statistical analysis.
- Served as data and M&E lead for UNICEF behavior change and awareness studies: tools, trackers, datasets, reach analysis, and report drafting.

Researcher / Data Scientist, Vanilla Value-Chain, January 2024 - June 2025

Makerere University / MAAIF, Catholic Relief Services, and sector partners, Uganda

- Researched vanilla maturity and harvest timing for national harvest-declaration considerations, combining field and crop work with statistical modeling and thermal-time analysis.
- Integrated field, satellite climate, and survey data and built time-series models to compare maturity-assessment methods and their timing, quality, and price implications.
- Conducted interviews with farmers and partners, field measurements, and seasonal maturity assessments across vanilla-growing districts.

- Helped establish the Kabanyolo demonstration plot and prepared models, scenario notes, and technical inputs for supervisors and project partners. Work concluded with program funding closure.

Research Associate, February 2023 - November 2023

Innovations for Poverty Action (IPA), Kampala, Uganda

- Supported development and implementation of process evaluation plans with program managers.
- Designed survey instruments and supported piloting, translation, and computer-assisted interviewing.
- Coordinated evaluation logistics including timelines, field staffing, and procurement, and managed data entry, cleaning, and analysis.
- Applied data-quality protocols including enumerator audits and monitoring of data collection.
- Maintained relationships with partner organizations, contributed to meetings and reports, and supported research approvals and compliance.

Qualitative Research Associate, July 2023 - March 2024

USAID Strategic Information Technical Support (SITES), Uganda

- Trained on field methods and tools and managed questionnaires and tablets for fieldwork across Eastern and Northern Uganda.
- Assessed facility readiness for Health Information Exchange and paperless viral load requests, reviewing UgandaEMR records against ART registers.
- Conducted key informant interviews and FGDs with clinical and data staff and checked data quality across HMIS tools and the EMR.
- Documented constraints affecting EMR use, including DHIS2 knowledge, internet and power stability, and computer performance, and shared facility-level feedback at each visit.
- Compiled progress and activity reports with safe backup and storage, completing a 12-day assessment in September 2023 within a seven-region exercise covering Kigezi, Ankole, Lango, Acholi, Busoga, Bugisu, and Karamoja.

Research Associate, September 2021 - January 2023

Bronkar (U) Limited, Kampala, Uganda

- Developed assignment workplans and tracked deliverables across research assignments.
- Designed methods and survey tools for qualitative and quantitative collection.
- Trained enumerators and participated in field research including interviews and structured surveys.
- Cleaned and analyzed survey data and prepared findings for client reports and proposals.

Research Assistant, April 2021 - June 2021

Makerere University Agricultural Research Institute, Wakiso, Uganda

- Collected data from field trials and cleaned data in Excel for analysis.

Data Entry Clerk, January 2021 - February 2021

UNYFA, Kampala, Uganda

- Entered and checked survey data for organizational databases and ran quality audits to keep records complete before analysis.

PORTFOLIO

- GitHub and portfolio: github.com/mugarrajohnson and johnsonmugarra.com, with repositories, project write-ups, live D3.js analytics, and publications with DOIs.
- Vanilla forecasting: Prophet and ARIMA models combining NASA POWER climate data with field measurements; interactive harvest model on the website, presented nationally at Makerere.
- Farm data platform (Enimiro): seven scheduled Python and R pipelines moving SurveyCTO submissions into PostgreSQL, feeding Power BI and Looker dashboards; available as a private walk-through on request.
- Automated evaluation pipeline: Python ingestion, cleaning, and report generation that makes the recurring evaluation cycle repeatable.

EDUCATION

M.Sc. Data Science, September 2022 - October 2024

Manipal Academy of Higher Education (MAHE), Karnataka, India

Coursework: Machine Learning, Deep Learning, Predictive Analytics, Time-Series Forecasting

B.Sc. Agriculture Science, August 2016 - May 2021

Makerere University, Kampala, Uganda

CERTIFICATIONS

- SAS Certified Statistical Business Analyst
- Microsoft Certified: Power BI Data Analyst Associate
- Tableau Desktop Specialist
- Google Project Management Certificate
- Human Subjects Research (CITI)

REFERENCE

- Available on request.